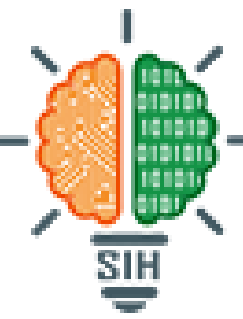


SMART INDIA HACKATHON 2025



SMART INDIA
HACKATHON
2025



AgroAura : AI-Powered Farming Assistant



- **Problem Statement ID - SIH25074**

- **Problem Statement Title -**

AI-Powered Personal Farming Assistant for Kerala Farmers

- **Theme - Agriculture , FoodTech & Rural Development**

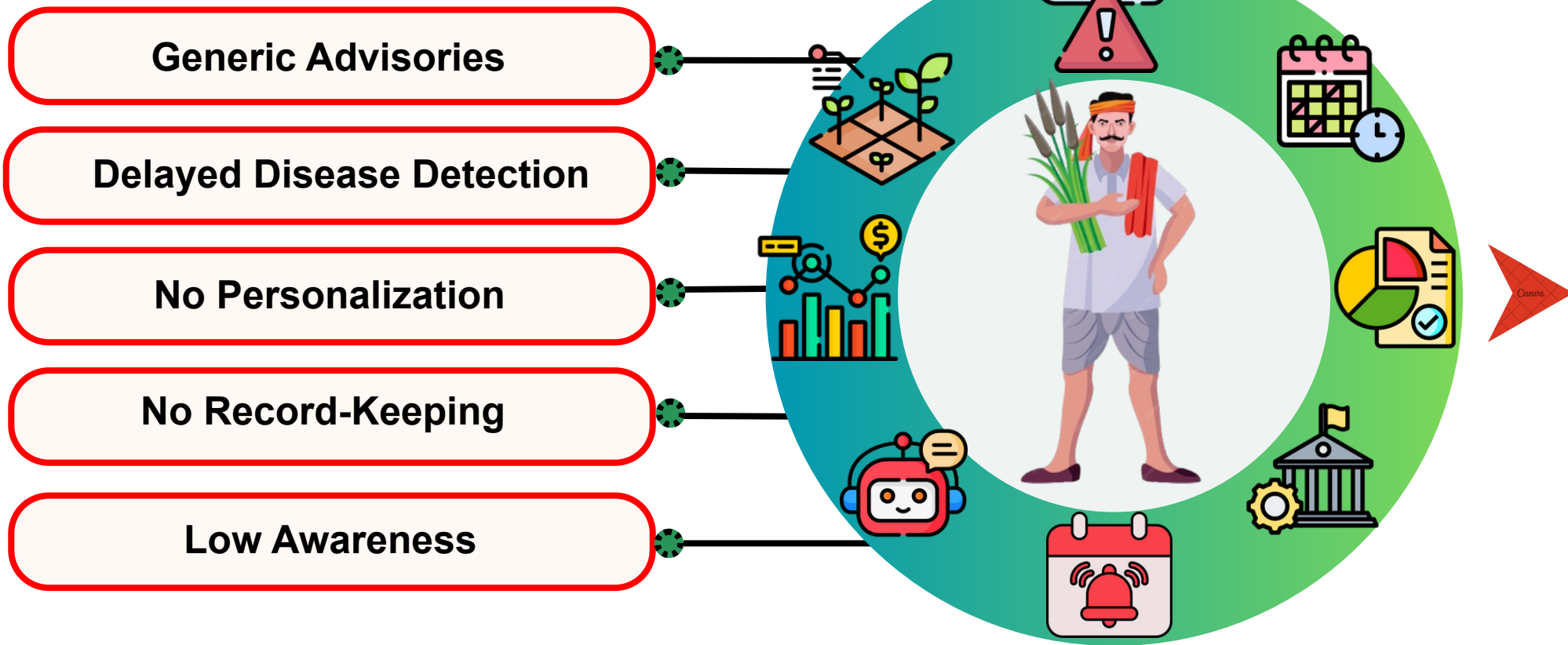
- **PS Category- Software**

- **Team ID-**

- **Team Name - Synapse**



FARMER CHALLENGES



UNIQUE VALUE PROPOSITION

FEATURES	Plantix	KissanAI	OUR SOLUTION
AI-Powered Crop Recommendation	NO	YES	YES
Bio-Digital Twin	NO	NO	YES
Kerala-Centric Design	NO	NO	YES
Integrated Marketplace for Farmers	NO	YES	YES
Voice Integrated ChatBot	NO	NO	YES
Offline-First with SMS + Report	NO	NO	YES
Government Scheme Guidance	NO	YES	YES
Disease Prediction & Treatment Tips	YES	NO	YES

PROPOSED SOLUTION

- AgroAura** is a digital friend providing **personalized, accessible, and real-time agricultural support.**
- ❖ **Crop Planner** → Suggests crops by soil, season & budget; shows cost, ROI, subsidy & govt. schemes, crop-specific suppliers
 - ❖ **Disease Detection** → AI diagnoses from photos; **early alerts** with organic & chemical remedies.
 - ❖ **Crop BioTwin** → A **virtual twin** growing ahead; gives **daily care steps** with simple visuals.
 - ❖ **Offline + SMS Alerts** → Works without internet; **auto-syncs** later. Sends SMS on tasks, weather & disasters.
 - ❖ **Voice Chatbot & Read-Aloud** → Farmers ask in Malayalam; app replies & reads instructions aloud.
 - ❖ **Market Analysis** → Compares mandi, co-op & buyer prices; shows **price trend graphs** & supports group selling.
 - ❖ **Report Generator** → Creates **harvest reports** (yield, income, photos) for **loans, insurance & subsidies.**
 - ❖ **Smart Agents** → **Planner, Risk, Market, Reminder & Scheme agents work together** like digital farming team.

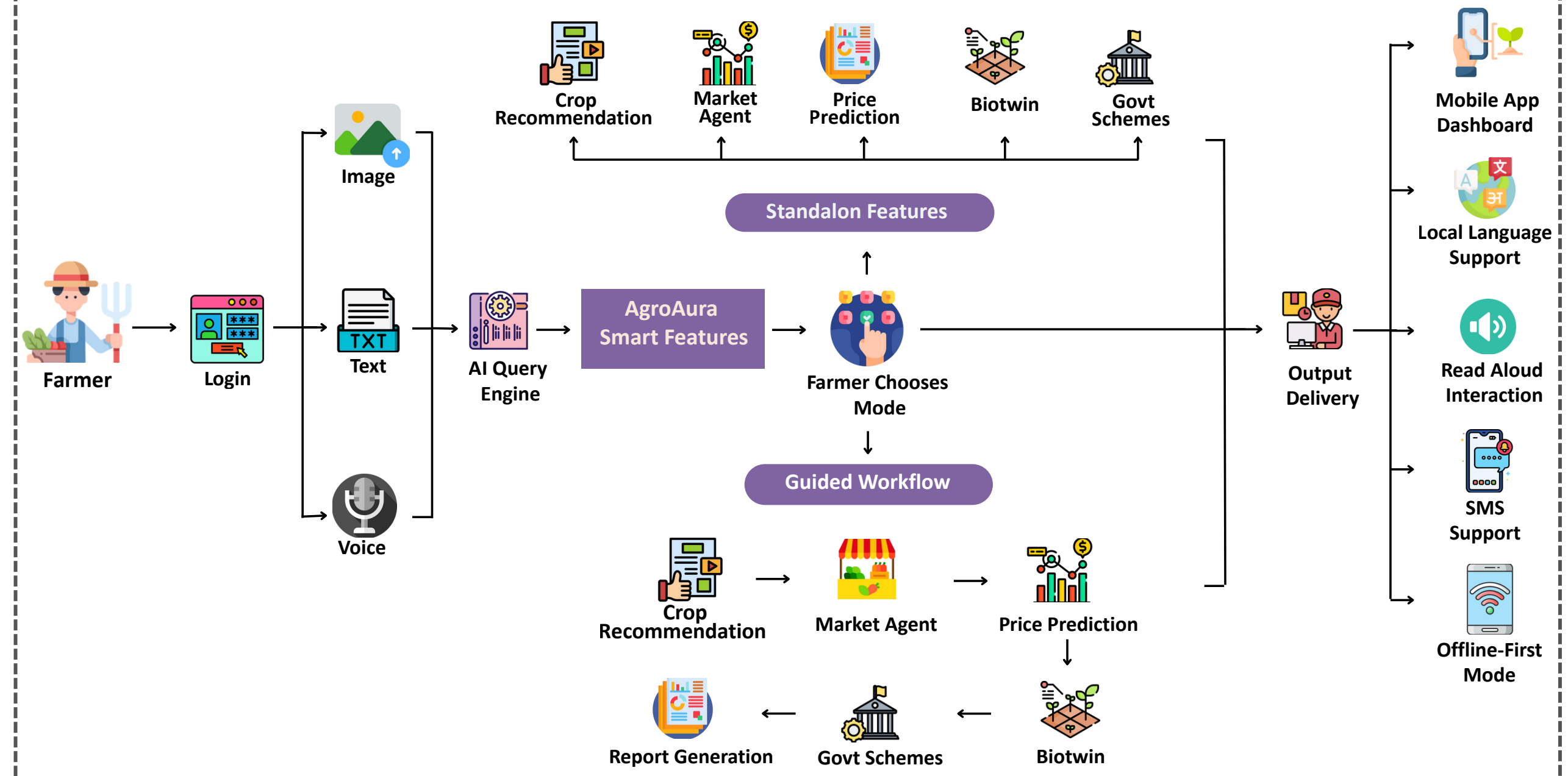
AgroAura Development



Tech Stack

- **React.js** : Interactive 3D visualization
- **HTML5, CSS3** : User Interface
- **Streamlit** : Farmer-friendly web dashboard
- **Python** : AI/ML models, backend logic, data processing
- **PyTorch & scikit-learn** : AI model like Random Forest, CNN, LSTM (crop recommendation, disease detection & price prediction)
- **Numpy+ SciPy + PIL** : Image analysis for crop/soil health
- **Transformers (HuggingFace)** : NLP for voice/text queries
- **Whisper ASR** : Speech-to-text voice input
- **Matplotlib** : Graphs & visualizations
- **Firestore** : Storing user info, farm inputs & session data
- **SQLite** : Offline storage + Auto-sync
- **Authentication & Encryption** : OTP-based login, HTTPS/TLS for API calls

Workflow Architecture - AgroAura



Prototype Link : <https://youtu.be/XUEdLU78g0o>

The product is **60% complete**, with our current focus on **developing the remaining features** and building a **sustainable revenue model**, while also **enhancing the UI** to ensure a more **farmer-friendly experience**.

FEASIBILITY AND VIABILITY

FEASIBILITY



Working Prototype: Core modules (Crop Planner, Disease Detection, BioTwin, Chatbot) already implemented.



Datasets & ML Model: Uses open datasets (PlantVillage, UCI) + Kerala-specific data (KAU, ICAR).



Offline-First: Local database with cloud sync; SMS/IVR support for non-smartphone users.



Easy Adoption: Simple web/mobile app; Malayalam voice & read-aloud; minimal training needed.



Scalable & Relevant: Kerala-focused now; modular Smart Agents make expansion to other regions possible.

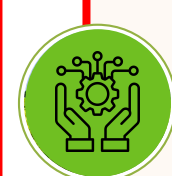
CHALLENGES



Language & Tech Limitations: Farmers face barriers with local language support & limited digital skills



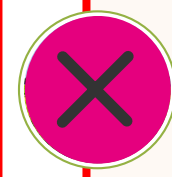
Connectivity Gaps: Poor connectivity in rural areas



Lack of Awareness : Farmers unaware of current practices



Integration Barriers: Difficulty linking with govt/agribiz systems



Wrong Info : AI may provide wrong information



STRATEGIES



Multilingual UI: Voice/text in 4+ languages



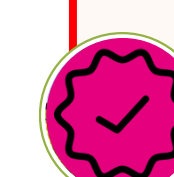
Offline + IVR: SMS/voice support for inclusivity



Virtual Demo : Shows a virtual demo of how Plant grows via Bio-twin!



Modular API: Easy plug-in with govt/agribiz systems



Verified Output: Recommendations are verified with Govt. bodies, Agri-universities, KVKs

BUSINESS APPROACH



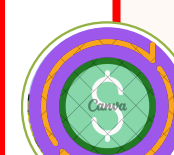
Freemium App : Free crop basics, paid pro features (disease AI, economics)



B2C: Low-cost subscription for farmers



B2B: Tie-ups with agri-banks, insurers, input suppliers



Data Monetization: Insights for markets, buyers, co-ops



PPP Model : Partner with Govt + AgriTech startups

IMPACT AND BENEFITS



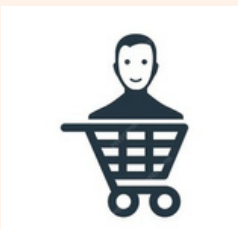
For Farmers

- Better crop yield & productivity
- Easy access to govt. schemes & market info



For Government

- Reliable agri-data for policymaking
- Contribution to national food security goals



For Consumer

- Access to healthier & safer food
- Stable prices through efficient supply chain



Social Benefits

- Empowers small & marginal farmers
- Inclusive & accessible (voice + local languages)



Environmental Benefits

- Reduces crop loss through early detection
- Optimizes water, fertilizer, and pesticide usage



Economic Benefits

- Increases yield & income for farmers
- Enhances market access & price prediction

SUSTAINABLE DEVELOPMENT GOALS



Zero Hunger

Enhances crop productivity and supports efficient farming to increase food security.



Industry, Innovation and Infrastructure

Implements smart technologies for farming, promoting innovation in agri-tech.



Decent Work and Economic Growth

Boosts rural employment by increasing farm efficiency and productivity.



Responsible Consumption and Production

Sustainable Management of Resources, Awareness and Education for Sustainability

Sno	Literature Survey	Publisher	Method Used	Works Done	Accuracy Achieved
1	Detection of Vegetation Encroachment in Power Transmission Line Corridor from Satellite Imagery Using Support Vector Machine: A Features Analysis Approach Fathi Mahdi Elsiddig Haroun et al	 Energies 2021	SVM	Built an AI-powered farming assistant for personalized crop planning and disease detection. Provided real-time, multilingual support using voice, photo, and chatbot with offline capabilities.	98.27%
2	Detection and Monitoring of Power Line Corridor From Satellite Imagery Using RetinaNet and K-Mean Clustering Norashidah et al.	 IEEE Access 2021	K-mean Clustering Algorithm	Developed AI models for crop planning, disease detection, and market prediction to help farmers make better decisions. Built a farmer-friendly assistant with smart agents, voice/NLP support, and a BioTwin for real-time guidance.	82.5%
3	A Review of Vegetation Encroachment Detection in Power Transmission Lines using Optical Sensing Satellite Imagery October 2020 Siti Noratiqah Mohamad Deros et al.	Power Transmission, 2020	CNN	Extended the work by adding crop planning, market prediction, and farmer support features beyond vegetation monitoring.	95%
4	Detection of Vegetation Areas From Satellite Images Roshan Shripad et al.	 IJSRT	CNN	Implemented CNN-based disease detection and extended it with crop planning, price prediction, multilingual chatbot, and BioTwin to build a complete smart farming system.	85%